

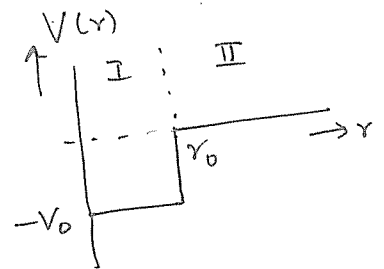
Solution set 11

Problem 19.5.4 (Shankar):

For s-wave scattering we need to solve.

$$\left\{ \frac{d^2}{dr^2} + \frac{2m}{\hbar^2} [E - V(r)] \right\} U(r) = 0$$

$$\begin{aligned} \text{We have } V(r) &= -V_0 & r < r_0 \\ &= 0 & r > r_0 \end{aligned}$$



$$\text{Thus if we define } \frac{2m(E+V_0)}{\hbar^2} = k'^2, \quad \frac{2mE}{\hbar^2} = k^2$$

we see that for

$$r < r_0 \quad U_{\text{I}}(r) \text{ obeys}$$

$$\left(\frac{d^2}{dr^2} + k'^2 \right) U_{\text{I}}(r) = 0$$

$$r > r_0 \quad U(r) \text{ obeys}$$

$$\left(\frac{d^2}{dr^2} + k^2 \right) U(r) = 0$$

Region I

$$U(r) = A' \sin k' r + B' \cos k' r$$

Region II

$$\begin{aligned} U(r) &= A \sin kr + B \cos kr \\ &= R \sin(kr + \delta_0) \end{aligned}$$

→ Wave function must vanish at $r=0$. $\Rightarrow B'=0$.

→ Wave function must be continuous at $r=r_0$.

$$A \sin k' r_0 = R \sin(k r_0 + \delta_0)$$

→ First derivative of the wave function must be continuous at r_0 .

$$A k' \cos k' r_0 = R k \cos(k r_0 + \delta_0)$$

∴ We see that

$$k \tan k' r_0 = k' \tan(k r_0 + \delta_0)$$

$$\therefore \delta_0 = -k r_0 + \tan^{-1} \left(\frac{k}{k'} \tan k' r_0 \right)$$

Neglecting $k r_0$ we see that-

$$\delta_0 = \tan^{-1} \left(\frac{k}{k'} \tan k' r_0 \right)$$

When $k' \approx k'_n = (2n+1)\pi/2r_0$

$$\text{then } \tan k'_n r_0 = \tan((2n+1)\pi/2) = \infty$$

$$\text{Thus } \tan(k' r_0) \approx \frac{1}{(k'_n - k') r_0} = \frac{k'_n + k'}{(k_n'^2 - k'^2) r_0}$$

$$= \frac{\frac{\hbar^2}{2m r_0} (k'_n + k')}{(E_n - E)} \approx \frac{\frac{\hbar^2}{m r_0} k'_n}{(E_n - E)} = \frac{\pi/2}{(E_n - E)}$$

$$\therefore \frac{\pi}{2} = \frac{\hbar^2}{m r_0} k'_n$$

Let us now consider bound states with energy E . Then we can find a condition.
 if the bound states are of energy $E = -V_0$. Then we must

have

$$A \sin k' r_0 = R e^{-k r_0}$$

$$A k' \cos k' r_0 = -R k e^{-k r_0}$$

$$\therefore \tan k' r_0 = -\frac{k'}{k}$$

If $\sqrt{E \rightarrow 0}$ or $k \rightarrow 0$ then $k' r_0 = (2n+1)\pi/2 = k_n' r_0$

$\therefore k' = k_n'$ is the point where we get bound states of energy ≈ 0 !

Now.

$$\begin{cases} \frac{\hbar^2 k^2}{2m} = |E| \\ \frac{\hbar^2 k'^2}{2m} = (V_0 - |E|) \end{cases}$$

Problem 19.5.6 (Shankar).

(1) We know that the full wavefunction of scattering is

$$\begin{aligned} \psi_k(\vec{r}) &= e^{ikz} + f(\theta, \phi) \frac{e^{ikr}}{r} \\ &= e^{ikr \cos \theta} + f(\theta, \phi) \frac{e^{ikr}}{r} \end{aligned}$$

Now the probability current is

$$\begin{aligned} \vec{j} &= \frac{\hbar}{2mi} \left[\psi^* \vec{\nabla} \psi - (\vec{\nabla} \psi) \psi^* \right] \\ &= \frac{\hbar}{m} \text{Im} [\psi^* \vec{\nabla} \psi] \end{aligned}$$

$$\lim_{r \rightarrow \infty} \vec{j} = \frac{\hbar}{m} \left[\underbrace{k \hat{z}}_{\vec{j}_{\text{inc}}} + \underbrace{|f|^2 \frac{k}{r^2} \hat{r}}_{\vec{j}_{\text{sc}}} + \underbrace{\frac{1}{r} \text{Im} \{ i k e^{ikr(\cos \theta - 1)} f^* \hat{z} + i k e^{-ikr(\cos \theta - 1)} f \hat{r} \}}_{\vec{j}_{\text{interference}}} \right]$$

$$\lim_{r \rightarrow \infty} \vec{j}_{int} = \frac{\hbar k}{m} \text{Im} \left\{ i \frac{e^{ikr(\cos\theta-1)}}{r} f^*(\theta) + e^{-ikr(\cos\theta-1)} \frac{f(\theta)}{r} \right\}$$

↑
interference
term

The radial component is given by

$$\lim_{r \rightarrow \infty} j_{int, r} = \frac{\hbar k}{m} \frac{1}{r} \text{Im} \left[i e^{ikr(\cos\theta-1)} f^*(\theta) \cos\theta + i e^{-ikr(\cos\theta-1)} f(\theta) \right]$$

(2) When we integrate over a small solid angle at $\theta \neq 0$ we see $(\cos\theta - 1) \neq 0$. Hence $e^{ikr(\cos\theta-1)}$ oscillates rapidly ^{even} over the small integration region and gives zero!

(3) When $\theta \approx 0$ then $(\cos\theta - 1) \approx -\frac{\theta^2}{2}$. Then the integral survives! We now see that $\cos\theta \approx 1$ $\sin\theta d\theta \approx \theta d\theta$

$$\int_0^\theta e^{-ikr\theta^2/2} f^*(\theta) \theta d\theta = \frac{f^*(0)}{kr}$$

$$\int_0^\theta e^{ikr\theta^2/2} f(\theta) \theta d\theta = -\frac{f(0)}{kr}$$

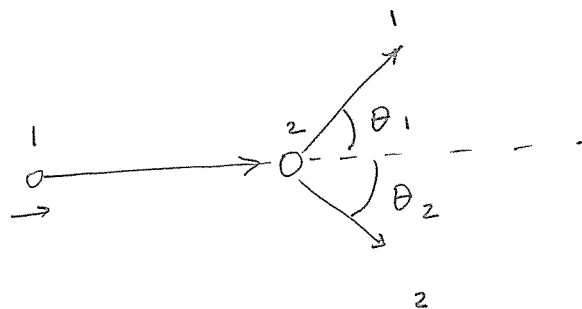
$$\begin{aligned} \int_{\text{forward cone}} j_r^{int} r^2 d\Omega &= \frac{\hbar k}{m} \frac{1}{r} \text{Im} \left[i \frac{f^*(0)}{kr} - \frac{i f(0)}{kr} \right] 2\pi r^2 \\ &= -\frac{\hbar k}{m} \left(\frac{4\pi}{k} \right) \cdot \text{Im} f(0) \end{aligned}$$

(2) . We can see that- $\theta_2 \leq \pi/2$ from energy - momentum conservation .

To see this note that-

$$mV = mV_1 \cos \theta_1 + mV_2 \cos \theta_2$$

$$mV_1 \sin \theta_1 = mV_2 \sin \theta_2$$



Clearly we can assume without loss of generality that-
 $\theta_1 > 0$ and $\theta_2 > 0$. (because the reverse situation $\theta_1 < 0$ and $\theta_2 < 0$ is a symmetric situation) .

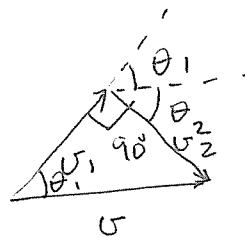
Thus ,

$$V = V_1 \cos \theta_1 + V_2 \cos \theta_2$$

$$V_1 \sin \theta_1 = V_2 \sin \theta_2$$

and energy conservation $\Rightarrow V^2 = V_1^2 + V_2^2$.

Geometrically



$\therefore V_1, V_2$ and V form sides of a right triangle .

$\Rightarrow \theta_1 + \theta_2 = \pi/2 \implies$ Along with $\theta_1 > 0, \theta_2 > 0$
means $\theta_1 \leq \pi/2$

$$\therefore \frac{\int_{\text{forward cone}} j_r^{\text{int}} r^2 d\Omega}{j_{\text{inc}}} = \underset{\substack{\uparrow \\ \text{deficit (depletion)}}}{-} \left(\frac{4\pi}{k} \right) \text{Im } f(0).$$

Now $\sigma_{\text{tot}} = \text{total scattering cross section.}$

$$\therefore \sigma_{\text{tot}} = \left(\frac{4\pi}{k} \right) \text{Im } f(0) \quad \Leftarrow \text{Optical theorem.}$$

Problem 19.6.1 (Shankar).

(1) We know that

$$\left(\frac{d\sigma}{d\Omega_L} \right) = \left(\frac{d\sigma}{d\Omega_{\text{cm}}} \right) \cdot \left(\frac{d\Omega_{\text{cm}}}{d\Omega_L} \right)$$

$$\text{Now } d\Omega_{\text{cm}} = \sin\theta_{\text{cm}} d\theta_{\text{cm}} d\phi_{\text{cm}}.$$

$$d\Omega_L = \sin\theta_L d\theta_L d\phi_L.$$

From eq. (19.6.16) we see that $2\theta_L = \theta_{\text{cm}} \quad \phi_L = \phi_{\text{cm}}$
6.17

$$\therefore \left(\frac{d\Omega_{\text{cm}}}{d\Omega_L} \right) = \frac{\sin 2\theta_L}{\sin \theta_L} \underbrace{\frac{d\theta_{\text{cm}}}{d\theta_L}}_2 \underbrace{\frac{d\phi_{\text{cm}}}{d\phi_L}}_1$$

$$= 4 \cos \theta_L$$

$$\left(\frac{d\sigma}{d\Omega_L} \right) = \left(\frac{d\sigma}{d\Omega_{\text{cm}}} \right)_{\theta_{\text{cm}} = 2\theta_L} 4 \cos \theta_L$$

(3) For unequal mass eqs. (19.6.14), (19.6.15) change as follows

$$\tan \theta = \frac{P'_\perp}{P'_z} \quad \tan \phi = \frac{P'_y}{P'_x} \quad (\text{same as } m_1 = m_2)$$

$$\tan \theta_L = \frac{P'_\perp}{\left(P'_z + \frac{m_1}{m_2} P\right)} \quad \tan \phi_L = \frac{P'_y}{P'_x} \quad \longrightarrow$$

$$\therefore \phi_L = \phi \quad \text{but} \quad \tan \theta_L = \frac{P'_\perp / P}{\left(\frac{P'_z}{P} + \frac{m_1}{m_2}\right)}$$

$$\text{Now } P'_\perp / P = \sin \theta \quad P'_z / P = \cos \theta$$

$$\therefore \tan \theta_L = \frac{\sin \theta}{\cos \theta + \frac{m_1}{m_2}}$$

Problem 19.6.2 (Shankar).

We start with eq. (19.3.16) for $f_c(\theta)$.

$$f_c(\theta) = \frac{-\gamma}{2k(\sin \theta/2)^2} e^{-i\gamma \ln \sin^2 \theta/2 + iA} \quad (A = \text{some const.}) \quad \text{purely real.}$$

$$\text{Now } \frac{d\sigma}{d\Omega} = |f(\theta, \phi) + f(\pi - \theta, \phi + \pi)|^2$$

$$f_c(\pi - \theta) = \frac{-\gamma}{2k(\cos \theta/2)^2} e^{-i\gamma \ln \cos^2 \theta/2 + A}$$

$$|f_c(\theta)|^2 = \frac{\gamma^2}{4k^2 \sin^4 \theta/2} \quad |f_c(\pi - \theta)|^2 = \frac{\gamma^2}{4k^2 \cos^4 \theta/2}$$

$$f_c^*(\theta) f_c(\pi - \theta) = \frac{\gamma^2}{4k^2 \sin^2 \theta/2 \cos^2 \theta/2} \cdot e^{i\gamma \ln \tan^2 \theta/2}$$

$$f_c(\theta) f_c^*(\pi - \theta) = \frac{\gamma^2}{4k^2 \sin^2 \theta/2 \cos^2 \theta/2} e^{-i\gamma \ln \tan^2 \theta/2}$$

$\therefore$

$$\frac{d\sigma}{d\Omega} = \frac{\gamma^2}{4k^2} \left[\frac{1}{\sin^4 \theta/2} + \frac{1}{\cos^4 \theta/2} + \frac{2 \cos(\gamma \ln \tan^2 \theta/2)}{\sin^2 \theta/2 \cos^2 \theta/2} \right]$$

$$\text{If } \gamma = \frac{e^2 \mu}{\hbar^2 k} \quad \frac{r}{k} = \frac{e^2}{2E} \quad \frac{\gamma^2}{4k^2} = \left(\frac{e^2}{4E} \right)^2$$

$$\therefore \frac{d\sigma}{d\Omega} = \left(\frac{e^2}{4E} \right)^2 \left[\frac{1}{\sin^4 \theta/2} + \frac{1}{\cos^4 \theta/2} + \frac{2 \cos(\gamma \ln \tan^2 \theta/2)}{\sin^2 \theta/2 \cos^2 \theta/2} \right]$$